

American cranberry (*Vaccinium macrocarpon*) extract affects human prostate cancer cell growth via cell cycle arrest by modulating expression of cell...



Prostate cancer is one of the most common cancers in the world, and its prevalence is expected to increase appreciably in the coming decades. As such, more research is necessary to understand the etiology, progression and possible preventative measures to delay or to stop the development of this disease. Recently, there has been interest in examining the effects of whole extracts from commonly harvested crops on the behaviour and progression of cancer. Here, we describe the effects of whole cranberry extract (WCE) on the behaviour of DU145 human prostate cancer cells in vitro. Following treatment of DU145 human prostate cancer cells with 10, 25 and 50 $\mu\text{g ml}^{-1}$ of WCE, respectively for 6 h, WCE significantly decreased the cellular viability of DU145 cells. WCE also decreased the proportion of cells in the G2-M phase of the cell cycle and increased the proportion of cells in the G1 phase of the cell cycle following treatment of cells with 25 and 50 $\mu\text{g ml}^{-1}$ treatment of WCE for 6 h. These alterations in cell cycle were associated with changes in cell cycle regulatory proteins and other cell cycle associated proteins. WCE decreased the expression of CDK4, cyclin A, cyclin B1, cyclin D1 and cyclin E, and increased the expression of p27. Changes in p16(INK4a) and pRBp107 protein expression levels also were evident, however, the changes noted in p16(INK4a) and pRBp107 protein expression levels were not statistically significant. These findings demonstrate that phytochemical extracts from the American cranberry (*Vaccinium macrocarpon*) can affect the behaviour of human prostate cancer cells in vitro and further support the potential health benefits associated with cranberries.